

Method of pose tracking by event camera using LED marker points

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Abstract – Using of vision sensors to achieve pose tracking has become a hot spot currently in the domain of computer vision. traditional vision sensors. However, the frame output of the conventional vision sensor and the redundant image data have a large delay and are highly dependent on the light intensity. There is a need for an attitude tracking technique that relies less on light and has a lower latency to improve system performance. Compared with traditional vision sensors, a new type of event camera is inspired by the neural network principle to develop event-based vision sensing technology, which is more robust in terms of delay and dependence on light intensity. We applied event processing algorithms to high-frequency flashing LED markers, including frequency filtering, clustering, and feature-point tracking. In the pose tracking of the event camera, the attitude is obtained from the PnP (perspective-n-point) algorithm, and a least-squares method is used to optimize the pose tracking. The experiment uses high-frequency LED as marking points, showing the comparison of event camera output in high and low light intensity environments, stability testing of frequency filtering clustering algorithm, and Z-axis data error of pose tracking within 5%. This method can be applied to attitude tracking in low light intensity environments.

Keywords – Event Camera; Pose Tracking; LED Marker Points

stream^[1]. They are especially suited for use in low-light-intensity environments and, to a large extent, can solve the latency problem by reading the data only when the light intensity of the pixel changes. At present, event cameras are used in low-latency ranging^[2], three-dimensional SLAM (Simultaneous Localization and Mapping)^[3], 3D reconstruction^[4], and attitude tracking^[5-6].

The backdrop of this work is to determine the pose by visually tracking the marker points, such as for the mutual positioning of a group of drones^[7] and in the landing of drones^[8]. Event cameras are based on a principle different from that of conventional cameras, and the output event stream contains massive events. A key issue to be resolved for the above applications is how to remove useless noise from the event stream and stabilize the output event data. In this paper, we propose to use frequency-filtering, clustering, and target-tracking methods to realize the event camera's tracking of the flashing LED marker points. In the pose tracking of the event camera, the attitude is obtained from the PnP (perspective-n-point) algorithm, and a least-squares method is used to optimize the pose tracking. Experimental results show that the proposed method can be used in low-light-intensity environments to obtain high-accuracy tracking results.

I. INTRODUCTION

The frame output of conventional vision sensors and data processing of cluttered images have considerable latency problems and are highly dependent on light intensity. Event cameras are a new type of sensor different from conventional cameras. They do not capture the intensity image at a fixed rate, but instead measure the change of the light intensity of a pixel asynchronously in the form of an event

II. EVENT CAMERA PRINCIPLE

A comparison of the operating principle of an event camera and a conventional CMOS (Complementary Metal-Oxide Semiconductor) camera is shown in Fig.1 A conventional CMOS camera records still frames of the high-frequency flashing sequence in a visual scene. There is no continuity between two frames and there is a time

period of a blind spot. During the blind-spot period, the algorithm is not operating.

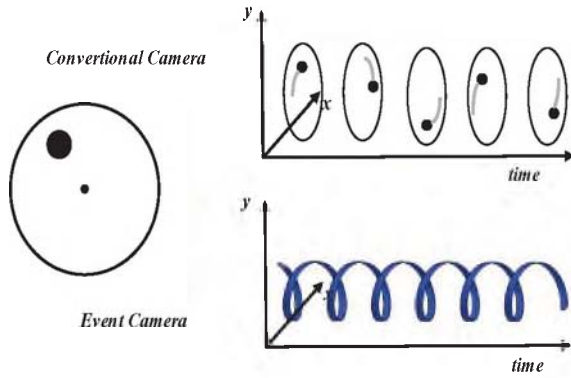


Fig.1 Event camera and conventional camera

An event camera only focuses on the light-intensity signal to be tracked and the data stream is continuous. This provides the obvious advantages of a very large dynamic range, no blurring due to motion, a high HDR(High Dynamic Range Imaging) ratio for weak as well as strong light, low latency (timing accuracy of approximately 1 μ s), and a low demand on bandwidth, i.e., only about one-tenth of that required by a conventional camera.

III. METHOD

A. Principle

The principle of using the event camera to track the marker points is shown in Fig.2 First, four marker points are set up in the world coordinate system: P_{w1} , P_{w2} , P_{w3} , P_{w4} . The event camera then captures the event and obtains four unordered feature points: D_1 , D_2 , D_3 , D_4 . Using the tracking algorithm, a one-to-one mapping is made from the four unordered feature points to the four marker points in the world coordinate system: P_{T1} , P_{T2} , P_{T3} , P_{T4} .

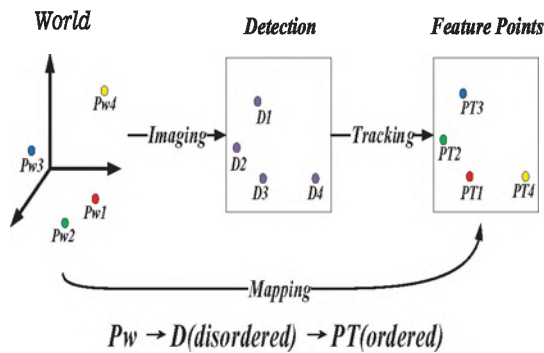


Fig 2. Principle of marker-point projection

Following the above principle, four high-frequency flashing LEDs are used as marking points. The signal output from the event camera is shown in Fig.3, in which a red dot represents an increase in light intensity and a blue dot a decrease in light intensity.

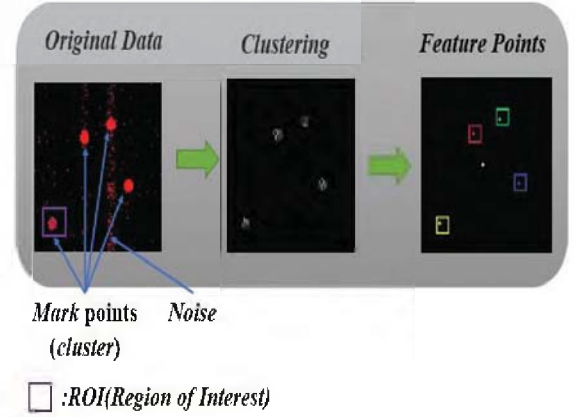


Fig.3 Event-processing steps

The event camera output data consist of a series of events; that is, pixel points of variable light intensity. Event e_i of the light-intensity change can be expressed as

$$e_i = \langle x_i(\text{pixel}), y_i(\text{pixel}), t_i, \text{Polarity}_i \rangle$$

where *Polarity* is the polarity of the pixel. When the light intensity increases, *Polarity*=True, and when the light intensity decreases, *Polarity*=False.

A Cluster is a data group produced by high-frequency flashing LED light. The region of interest (ROI) is a rectangular area containing clusters. The ROI is clustered to obtain the cluster center P (feature point) as the positions (D_1, D_2, D_3, D_4) of the marker point (high-frequency flashing LED light) in the image.

B. Implementation Principle

The method proposed in this paper is shown in Fig.2 Using the characteristics of high-frequency flashing LED lights, filtering is used to remove noise. Clusters of multiple high-frequency events appear after filtering, and each cluster is a set of high-frequency events generated by high-frequency flashing LEDs surrounded by the ROI. Clustering the ROI, one obtains the cluster cores, which are feature points. By tracking the feature points, one matches the feature points of the image to the world coordinate points one by one, and finally obtains the solution and optimizes the pose.

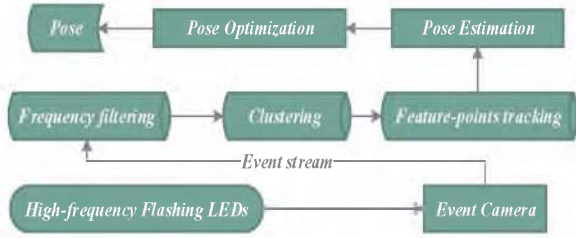


Fig.4 Flow diagram for pose tracking

a. Event-processing tracking

Event-processing tracking performs such operations as filtering, clustering, and feature-point tracking on the output data of the event camera. Fig.4 is the process, as described in the following subsections.

i. Frequency filtering

Frequency filtering filters low-frequency events (noise) and residual high-frequency events (high-frequency flashing lights). $E_{(n^*m)}$ is a set of events with pixel coordinates (n, m) . Each event e_i in the event stream is extracted into $E_{(n^*m)}$ according to the difference of pixel values:

$$E_{n^*m} = \{e_i \in E_{n^*m} | \text{impixelinfo}(e_i) = (n, m)\}$$

Within the same pixel coordinate, the frequency calculation is shown in Fig.5, where *Interval* is the transition time, and its value is equal to the current timestamp t minus the timestamp t' when the polarity was last changed, $\text{Interval} = t - t'$. *Frequency* is the frequency value of the pixel and $\text{Frequency} = 1/\text{Interval}$.

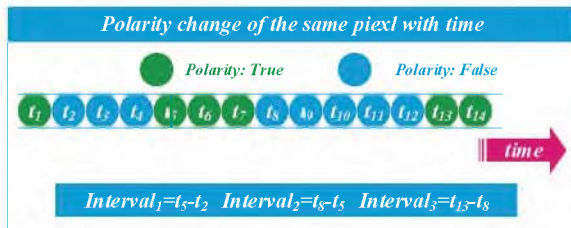


Fig.5 Flow diagram for frequency calculation

ii. Clustering

Within the ROI, the mean of the weighted pixels is calculated to obtain the cluster core P :

$$\bar{X} = \frac{\sum_{i=1}^n \text{Frequency}_i \cdot x_i}{n} \quad (1)$$

$$\bar{Y} = \frac{\sum_{i=1}^n \text{Frequency}_i \cdot y_i}{n} \quad (2)$$

$$P = (\bar{X}, \bar{Y}) \quad (3)$$

where n is the number of high-frequency events in a ROI, and Frequency_i is the frequency of the corresponding point $i (x_i, y_i)$.

iii. Feature-point tracking

When feature-point tracking establishes the correspondence between the feature points (cluster center) and the marker points of the world coordinate system, then the correspondence between (D_1, D_2, D_3, D_4) and $(P_{T1}, P_{T2}, P_{T3}, P_{T4})$ is realized. An example of a feature-point tracking method is shown in Fig.6, which is described as follows.

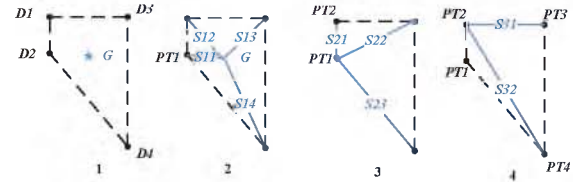


Fig.6 Schematic of feature-point tracking

(1) D_1, D_2, D_3, D_4 are connected head-to-tail to form a polygon and the centroid G is calculated as follows:

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n} \quad (4)$$

$$\bar{Y} = \frac{\sum_{i=1}^n y_i}{n} \quad (5)$$

$$G = (\bar{X}, \bar{Y}) \quad (6)$$

Here, $n = 4$ is the number of feature points and G is the centroid of the polygon.

(2) Calculate the distance between the four feature points and the geometric centroid G , and designate the feature point S_{11} closest to G as PT_1 .

(3) Calculate the distance between the remaining three feature points and PT_1 , and designate the feature point closest to PT_1 as PT_2 .

(4) Calculate the distance between the remaining two feature points and PT_2 , and designate the feature point closest to PT_2 as PT_3 .

(5) The last remaining point is PT_4 .

b. Pose estimation and optimization

i. Pose estimation

The pose estimation is carried out using the PnP algorithm[9]. The position of camera movement is constrained by the matching point of the pixel coordinate system and the world coordinate system. Combined with camera parameters, one obtains the estimated pose of the camera.

The algorithms for the parameters are the following.

(1) Input parameter:

Internal parameter of camera:

$$K = \begin{bmatrix} fx & 0 & cx \\ 0 & fy & cy \\ 0 & 0 & 1 \end{bmatrix} \quad (7)$$

Distortion coefficient:

$$D = [k_1 \quad k_2 \quad t_1 \quad t_2 \quad k_3] \quad (8)$$

Point in world coordinate system:

$$P_w = n * \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (9)$$

Point in pixel coordinate system:

$$P_T = n * \begin{bmatrix} u \\ v \end{bmatrix} \quad (10)$$

The value of n is determined by the PnP algorithm and, generally, $n \geq 4$.

(2) Output parameter:

Rotation matrix R is for the tri-axial rotational pose.

$$R = \begin{bmatrix} r_1 & r_2 & r_3 \\ r_4 & r_5 & r_6 \\ r_7 & r_8 & r_9 \end{bmatrix} \quad (11)$$

translational vector $\vec{t}$ is for the position.

$$\vec{t} = \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} \quad (12)$$

ii. Pose optimization

Pose optimization is performed using graphical optimization based on the least-squares method, with the goal being the minimization of re-projection error. The model consists of nodes and edges.

Node: The pose of the camera is $\xi \in \text{se}(3)$, and the 3D spatial position of all feature points is $X_j \in R^3$.

Edge: An edge is formed by projecting a 3D point onto the camera image, and is described by the observation equation: $z_j = h(P_i, X_j)$.

Assuming that a 3D point in space with coordinates $\mathbf{X}_i = [x_i, y_i, z_i]^T$ is projected to $\mathbf{u}_i = [u_i, v_i]^T$ in pixel coordinates, then their relationship is

$$s_i \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = K * \exp(\xi^\wedge) * \begin{bmatrix} x_i \\ y_i \\ z_i \\ 1 \end{bmatrix} \quad (13)$$

Expressed in matrix form, it is

$$s_i * \mathbf{u}_i = K * \exp(\xi^\wedge) * \mathbf{X}_i \quad (14)$$

One can construct a least-squares problem by finding the sum of the errors, and then finding the optimal pose of the camera and minimizing it:

$$\xi^* = \arg \min_{\xi} \frac{1}{2} \sum_{i=1}^n \left\| \mathbf{u}_i - \frac{1}{s_i} K \exp(\xi^\wedge) \mathbf{X}_i \right\|_2^2 \quad (15)$$

Solving the problem with the Gauss-Newton optimization method[10], one arrives at the following linear relationship:

$$e(P + \Delta P) \approx e(P) + J \Delta P \quad (16)$$

Here, $e \in R^{2*2}$ is the pixel coordinate error, $P \in R^{6*6}$ the camera pose, and $J \in R^{2*6}$ the Jacobi matrix.

IV. EXPERIMENT

The experimental system is shown in Fig.7 Four high-frequency flashing LED lights (1000 Hz or higher) are placed on a cardboard as markers. The lights are controlled by an Arduino¹ microcomputer with logic analysis capability. The cardboard is placed on the ground and remains stationary. The event camera is a model edvs-4337 unit in the Dynamic Vision Sensor (DVS) [11] series with a pixel value of 128×128.(inivation², ETH Zurich³)

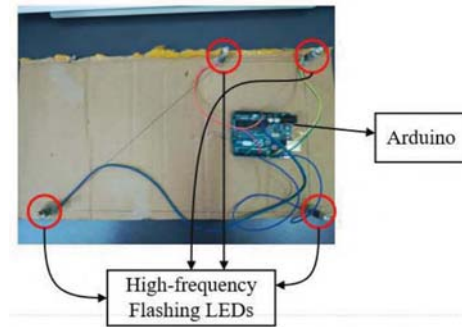


Fig.7 Cardboard with markers

A. Effects of light intensity

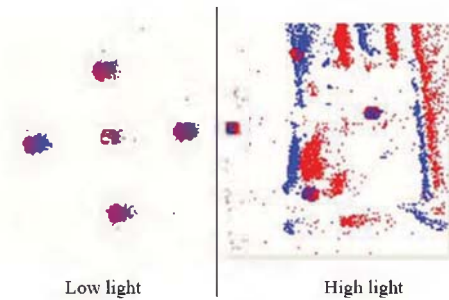


Fig.8 DVS camera moving in low- and high-light-intensity environment

The event camera is used to detect the marker points experimentally under dynamic environments of low and high light intensity. The results are shown in Fig.8 The LED board is flashed at a fixed high frequency and the event camera moved. There are noticeably fewer noise data at low light intensity and

the detected points are very distinct. There are more peripheral noise data at high light intensity. Not all the noise can be removed by filtering, and filtering may even interfere with the marker-point data. The test results show that the algorithm proposed here is more suited for a low-light-intensity environment and that new algorithms or better parameters are needed for high-light-intensity environments.

B. Stability test for clustering algorithm

The stability of the filtering and clustering algorithms is tested in a static environment. The relative position of the marker points and the event camera is fixed in low- and high-light-intensity environments. Data are taken randomly at 10 different times and the coordinates and their mean values estimated after clustering the marker points. The results are shown in Fig.9 and 10.

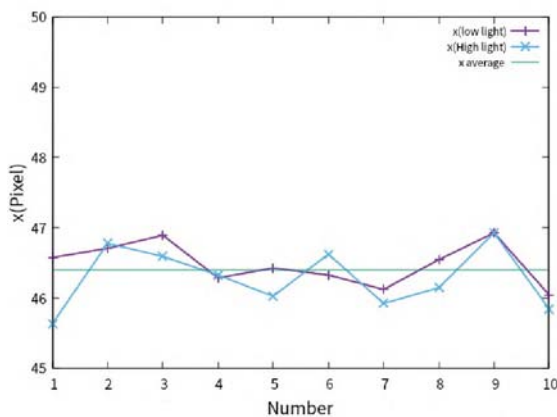


Fig.9 Estimated results for X coordinate

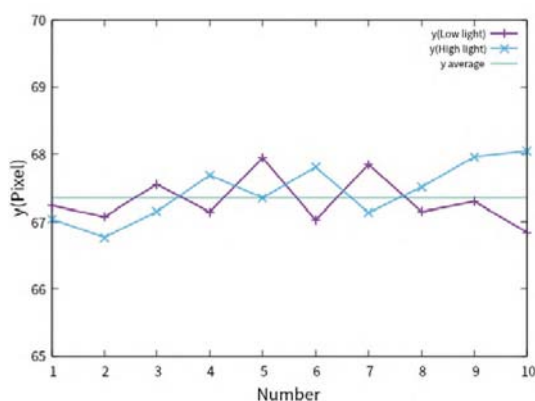


Fig.10 Estimated results for Y coordinate

Observation of the results shows that the difference between the estimated cluster center (x , y) and the mean value of the same marker point is within the range of $(-1 \sim 1)$ pixel. This shows that frequency filtering and clustering algorithms have good stability

in event processing.

C. Comparison of estimated height

In a low-light-intensity environment, the cardboard is fixed, with the high-frequency flashing LED lights on the ground and the event camera is moved. The program is run in the ROS environment, the marker-point data captured and transferred to the computer to calculate the current pose. The Z-axis data (height) and the actually measured height for the chosen position are compared. The actual heights chosen in the test are in the range 160–320 cm and the experimental data are shown in Fig.11 Error-analysis results, shown in Fig.15, display the $\text{error\%} = (\text{estimated height} - \text{actual height}) / \text{actual height}$. As can be seen from Fig.12, the range of error is less than 5% and the measurement data have good accuracy.

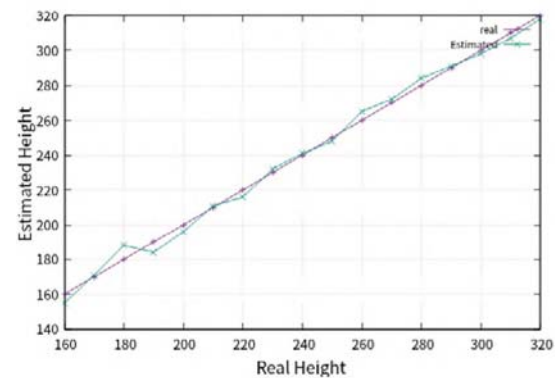


Fig.11 Comparison of Z-axis data with actual values

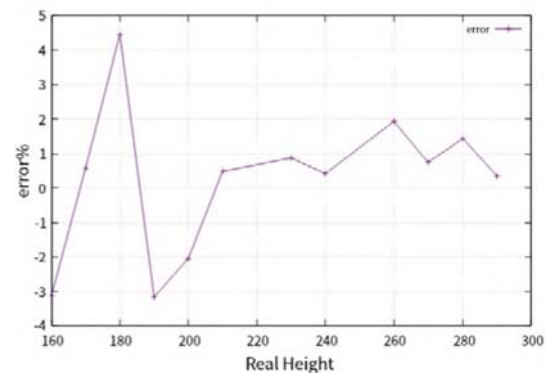


Fig.12 Error-analysis results diagram

V. CONCLUSIONS

In this paper, a frequency-based clustering and tracking algorithm to process data acquired by event camera is proposed, and then applied to pose

estimation. In terms of latency and the effects of light intensity, event cameras are more robust than conventional cameras for visually tracking pose. The event camera had lower noise and performed better in a low-light-intensity environment. The Z-axis error of the pose tracked by the method proposed in this paper was within 5%, so the method may be used to assist fixed-point landings of unmanned aerial vehicles (UAVs) by acquiring the altitude of the UAV in a low-light-intensity environment.

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